OFFICIAL OPERATIONAL MANUAL | SYSCOHADA REVISED (AUDCIF) | IFRS DUAL-PRESENTATION | HEALTHCARE CMMS & SAFETY

LedgerCore Health — Comprehensive User Guide & Operational Manual

Hospital Core Accounting, Asset Lifecycle & Maintenance Management System
Dual-Standard SYSCOHADA Revised (AUDCIF) & IFRS Compliance

1. System Overview & Architecture



Figure 1: LedgerCore Health Dual-Standard Architecture — Unified Double-Entry Core connecting SYSCOHADA Revised (OHADA), IFRS Presentation, Biomedical Asset Lifecycle, and Hospital Cash Desks.

1.1 Purpose & Mission

LedgerCore Health is an enterprise-grade Core Accounting, Healthcare Asset Lifecycle, and Clinical Engineering Management (CMMS) software system engineered specifically for hospitals, specialized clinics, and regional health networks operating in the **OHADA (*Organization for the Harmonization of Business Law in Africa*)** zone, CEMAC/UEMOA regions, and international healthcare development settings.

1.2 Dual-Framework Transaction Architecture

Unlike legacy enterprise systems that require maintaining parallel databases or duplicate data entries, LedgerCore utilizes a **unified double-entry transactional core**: - **Statutory SYSCOHADA Presentation:** Directly complies with the OHADA Uniform Act on Accounting and Financial Reporting (*AUDCIF*), generating regional tax packets, the 18-column Accounting Entries Audit File (FEC), and statutory annual financial statements (*Balance Sheet, Income Statement, Statement of Cash Flows*). -

International IFRS Presentation: Evaluates the identical underlying transactions against IFRS principles (such as IFRS 16 lease capitalization, economic useful life component depreciation, and fair value adjustments). - **Dual-Standard Bridge:** Quantifies and explains every variance between the statutory net surplus and international profit with zero unexplained discrepancies.

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2. Getting Started & Authentication

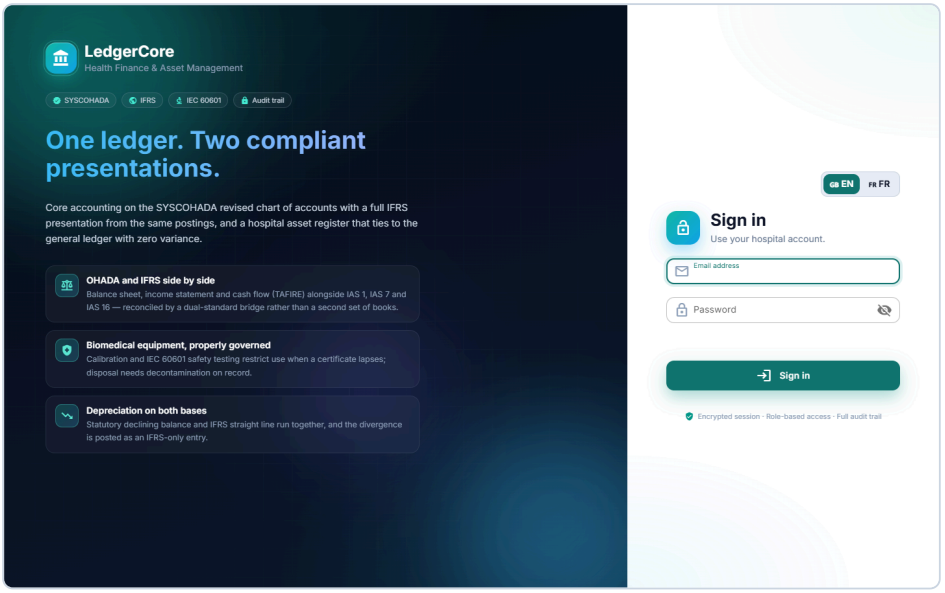


Figure 2: LedgerCore Authentication Interface — Dual-Language Selector (English / French), Encrypted Session Handshake, and Secure Sign-In Portal.

2.1 Accessing the Application

- 🌐 **Web URL:** `http://localhost:8700` (or your hospital's secure domain, e.g. `https://ledgercore.shisong.health`).
- ⚡ **Backend API Endpoint:** `http://localhost:4700/api`.
- 💻 **Supported Browsers:** Google Chrome, Microsoft Edge, Mozilla Firefox, Safari.

2.2 Step-by-Step Procedure: Logging into the System

1. 🌐 **Open Web Browser:** Navigate to the system URL.
2. 🌍 **Select Interface Language:**
   - In the top right corner of the authentication card, click **English** or **Français**.
   - *Result:* All labels, prompts, account classifications, and statutory terms instantly switch to the chosen language.
3. 🔑 **Enter User Credentials:**
   - In the **Email address** field, enter your assigned hospital email (e.g., `grace.ntamack@hgy.cm`).
   - In the **Password** field, enter your password.
   - Click the **Sign in** button.
4. 🛡️ **Role-Based Access Enforcement:**

- Upon successful credential verification, the system securely generates your session token and routes you directly to the workspace appropriate for your hospital role (Chief Financial Officer, Accounting Clerk, Biomedical & Asset Manager, Field Technician, or System Administrator).

5. 🔍 **Inspect Top Header Bar Indicators:**
   - Verify that your organization profile displays at the top:
     - Entity Name: `🏥 TSSF Shisong SOA`
     - Statutory Standard: `CM CM · 🟨 OHADA Normal System · 💱 XAF`
     - Quick Access Buttons: User Guide (`help_outline`), Print (`print`), and Reload (`refresh`).

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3. Module 1: Executive Dashboard & Operational KPIs

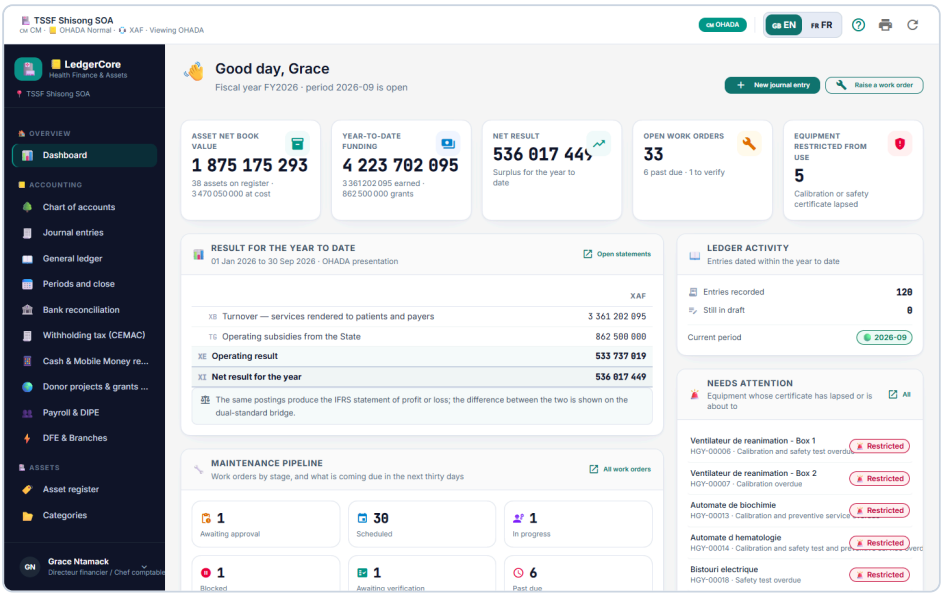


Figure 3: Executive Dashboard & Operational Pulse — Real-Time Net Surplus, Asset Net Book Value (NBV), Work Order Pipeline, and Clinical Quarantine Restriction Alert.

3.1 Purpose & Scope

The Executive Dashboard provides hospital leadership, finance directors, and biomedical heads with an instant real-time operational pulse of revenue, asset values, work order pipelines, and clinical safety compliance.

3.2 Step-by-Step Procedure: Reviewing Executive KPIs

1. 📊 **Navigate to Dashboard:** Click **Overview → Dashboard** (`/`) from the left navigation rail.
2. 💰 **Audit Financial Performance Cards:**
   - **Net Surplus / Deficit:** Shows year-to-date operating surplus or deficit calculated from posted revenue and expense lines.
   - **Operating Turnover:** Displays patient consultation fees, surgical fees, laboratory revenues, and pharmacy sales.
   - **Operating Subsidies & Grants:** Displays State operational subsidies and recognized donor contributions.

3. 🏷️ **Inspect Asset Health & Net Book Value:**

- **Asset Net Book Value (NBV):** Displays total capitalized gross cost minus cumulative depreciation.
- Click **Open register** on the asset card to drill down into category-level gross book values.

4. 🩺 **Audit Maintenance & Biomedical Safety Pipeline:**
   - Review open work orders grouped by status: Awaiting Approval, Scheduled, In Progress, Blocked, Awaiting Verification.
   - **Restricted / Quarantined Equipment Alert:**
     - 🟢 **Green Badge:** *"All equipment holds a current certificate"* — clinical assets are fully compliant.
     - 🔴 **Red Alert Shield:** *"Equipment restricted from use"* — one or more life-support or diagnostic devices have lapsed calibration or failed electrical safety tests. Click the badge immediately to review the quarantined assets.

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4. Module 2: Chart of Accounts & Analytical Structure

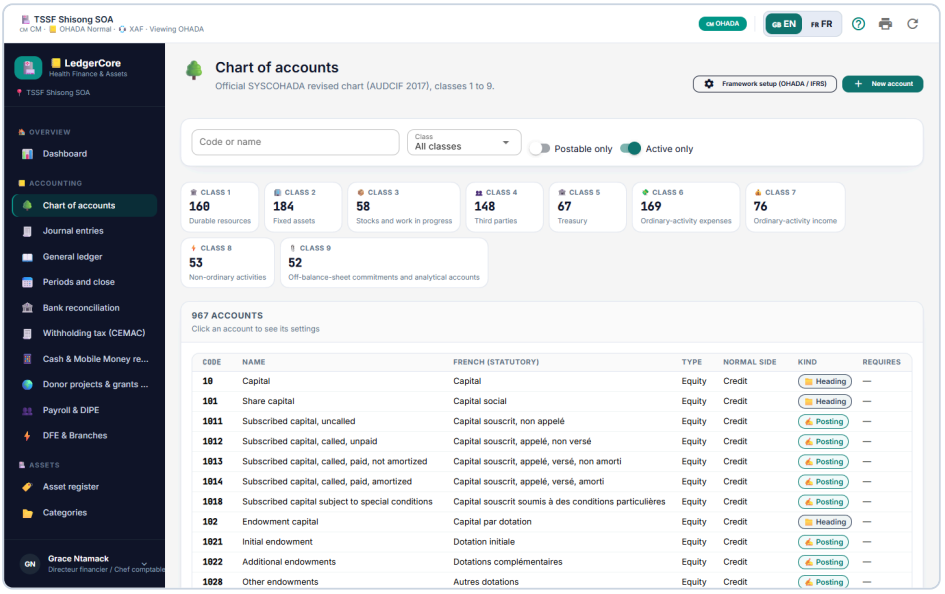


Figure 4: SYSCOHADA Revised Chart of Accounts — Class 1 to 9 Hierarchical Tree with Official Account Titles, Analytical Cost Centers, and 1-to-1 IFRS Presentation Alignment.

4.1 Purpose & Framework Rules

The Chart of Accounts module provides the statutory backbone of the hospital, implementing the **SYSCOHADA Revised** 9-class chart of accounts with hierarchical parent-child relationships and 1-to-1 IFRS presentation mapping.

4.2 Step-by-Step Procedure: Creating a New Postable Account

1. 📁 **Navigate to Accounts:** Click **Accounting → Chart of accounts** (`/accounts`).
2. 🔍 **Filter by Class or Keyword:** Select a class tab (e.g., **Class 6 — Expenses**) or search by code (e.g., `6011`).
3. ➕ **Initiate Account Creation:** Click the **+ New Account** button.
4. 📝 **Fill Account Parameters:**

- **Account Code:** Must start with the class digit and follow its parent (e.g., `601105`).
- **Official Account Name:** Statutory account title (e.g., `Surgical Consumables & Supplies`).
- **Parent Account:** Select the parent 3-digit or 4-digit heading (e.g., `6011`).
- **Accepts Postings (Switch):** Enable to allow transactional entries.
- **Requires Cost Centre (Switch):** Enable for clinical expenses requiring ward/department allocation.
- **Requires Third Party (Switch):** Enable for supplier or client collective accounts (`401`, `411`).
- **Requires Asset Reference (Switch):** Enable for maintenance and equipment accounts (`624`, `605`).

5. 🌐 **Set IFRS Presentation Line:** Choose the corresponding international reporting line (e.g., `Operating Expenses - Medical Supplies`).
6. 💾 **Save Account:** Click **Save Account**. The system verifies account code length and registers the account.

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5. Module 3: Journal Entries & AUDCIF Statutory Capture

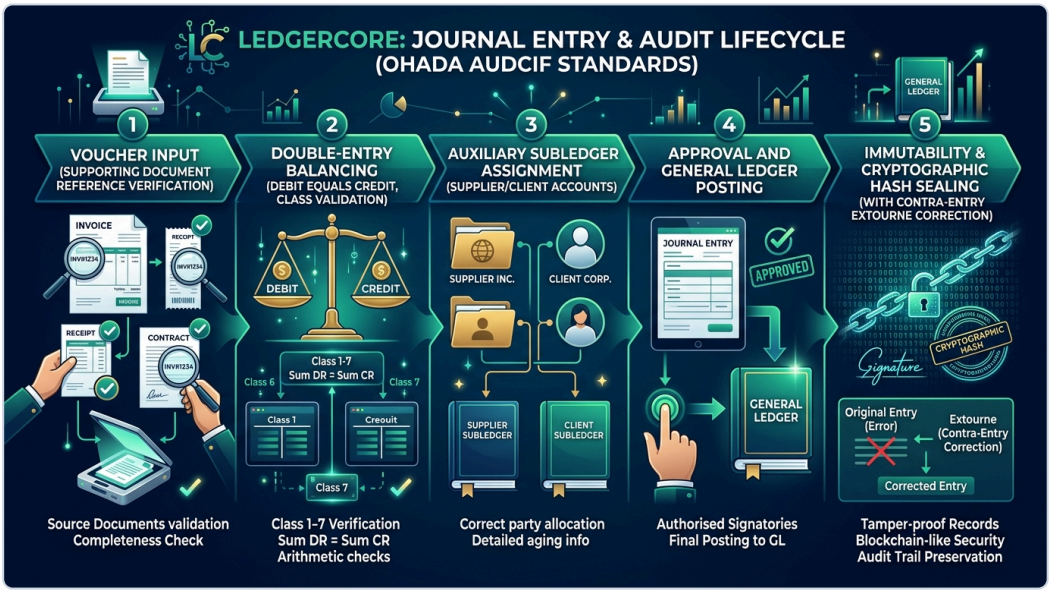


Figure 5: LedgerCore Journal Entry & Audit Lifecycle under OHADA AUDCIF Standards — From Voucher Input and Double-Entry Balancing to Auxiliary Subledger Allocation, General Ledger Posting, and Tamper-Proof Cryptographic Hash Sealing.

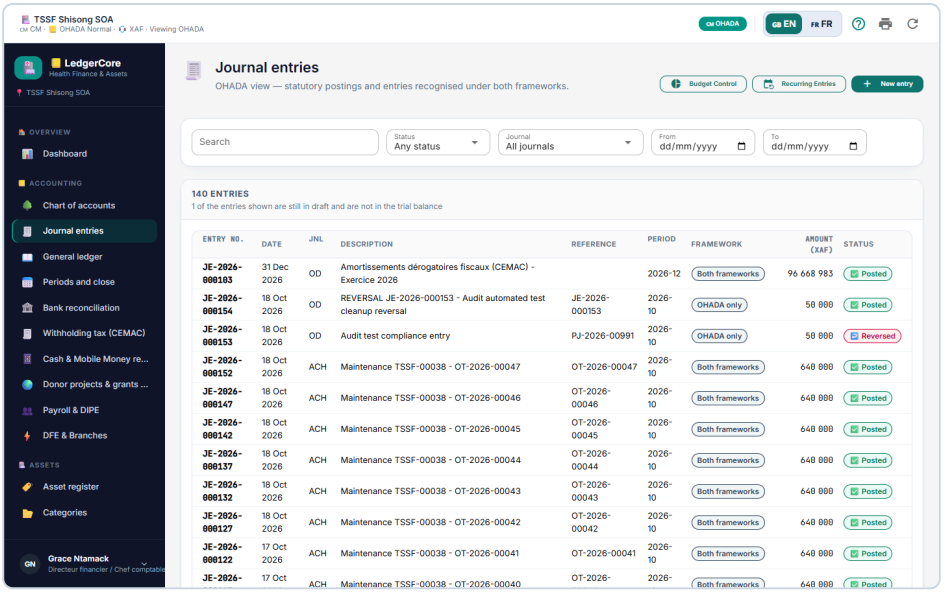


Figure 6: Journal Entries Ledger Screen — Chronological List of Vouchers with Reversal Links, Balanced Debits/Credits, and Real-Time Draft Count.

5.1 Purpose & Legal Controls

In strict compliance with **AUDCIF Articles 17, 19, and 20**: - Every entry must balance: $\sum \text{Debits} = \sum \text{Credits}$. - Every manual entry must cite a verifiable supporting voucher reference (*source document identifier*). - Negative numbers are strictly prohibited. - Posted entries cannot be edited or deleted; corrections are handled exclusively via contra-entry reversal.

5.2 Step-by-Step Procedure: Capturing a Double-Entry Voucher

1. 📝 **Navigate to Journal Entries:** Click **Accounting → Journal entries** (`/journal`).
2. ➕ **Open Capture Modal:** Click the **+ New Entry** button.
3. 📚 **Select Specialized Journal:**
   - `ACH` (Purchases & vendor bills)
   - `VTE` (Patient consultations & medical sales)
   - `BQ` (Bank transactions)
   - `CA` (Cash desk & Mobile Money)
   - `OD` (General accruals & adjustments)
4. 🗓️ **Set Accounting Date:** In **Entry date**, enter the transaction date (must fall in an open period).
5. 🏷️ **Enter Mandatory Supporting Voucher Reference (AUDCIF Art. 17):**
   - In **Reference ***, type the invoice number, receipt number, or bank slip ID (e.g., `FAC-2026-9901`). Leaving this blank triggers an immediate validation rejection.
6. 📄 **Enter Header Description:** Provide an operational summary (e.g., `Purchase of Laboratory Reagents - BioMed Services`).
7. ✍️ **Populate Journal Lines (Minimum 2 Lines):**
   - **Line 1 (Debit):**
     - Select **Account** (e.g., `601100 — Medical Supplies`).
     - Enter **Debit Amount** (e.g., `150,000.00`).
     - Select **Cost Centre** (e.g., `Medical Laboratory`).

- **Line 2 (Credit):**
  - Select **Account** (e.g., `401100 – Trade Payables / Suppliers`).
  - Enter **Credit Amount** (e.g., `150,000.00`).
  - Under **Third Party**, choose the specific vendor (e.g., `[FRN] BioMed Services Africa`).

8. ⚖️ **Verify Balance Indicator:**
   - Check the **Out of Balance** counter at the bottom right. When balanced, it displays `0.00` in green.

9. 💾 **Save Voucher:**
   - Click **Save as Draft** to queue for chief accountant review.
   - Click **Save and Post** (if holding `journal.post` permission) to commit lines to the General Ledger immediately.

5.3 Step-by-Step Procedure: Reversing a Posted Entry (Contra-Entry)

1. 🔍 Locate the posted entry in the journal list and click to open its detail modal.
2. ↩️ In the footer, click the **Reverse Entry** button (`undo` icon).
3. 📝 In the reversal dialog:
   - Verify the **Reversal Date** (defaults to today's date).
   - In **Reason for Reversal**, enter an audit justification of at least 5 characters (e.g., `Correction of third-party account allocation error`).
4. 🚀 Click **Reverse entry**. LedgerCore creates an inverted contra-entry, links it to the original voucher, and commits it with a distinct sequential entry number.

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6. Module 4: General Ledger, Auxiliary Subledgers & Open-Item Matching

6.1 Purpose & Scope

The General Ledger provides complete chronological running balances, while auxiliary subledgers maintain itemized partner accounts for suppliers, patients, and insurers.

6.2 Step-by-Step Procedure: Generating the 8-Column Trial Balance

1. 📖 Navigate to **Accounting → General Ledger** (`/ledger`).
2. 📊 Click the **Trial Balance** tab.
3. 🗓️ Select the reporting date range (**From** and **To**).
4. 🌐 Select the **Framework** (`OHADA` or `IFRS`).
5. 🔍 Click **Apply Range**. The system generates the statutory 8-column table:
   - *Columns 1 & 2:* Opening Debit & Opening Credit
   - *Columns 3 & 4:* Period Movement Debit & Period Movement Credit
   - *Columns 5 & 6:* Year-to-Date Cumulative Debit & Credit
   - *Columns 7 & 8:* Closing Debit & Closing Credit

6. 📥 Verify that total debits equal total credits across all column pairs. Click **Excel** or **PDF** to export.

6.3 Step-by-Step Procedure: Executing Open-Item Matching (Account Reconciliation)

1. 👤 In the General Ledger, select an auxiliary partner account (e.g., `401100 - Trade Payables / Suppliers`).
2. 🔍 Set the **Matching Filter** to **Unmatched Items**.
3. ☑️ Locate the original supplier bill credit entry and the corresponding bank payment debit entry.
4. ⚖️ Check the selection boxes next to both entries. Verify the net difference between selected debits and credits is zero.
5. 🔤 Click **Match Selected Items**. The system assigns a matching code (e.g., `AA`) and locks the matched pair.

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7. Module 5: Accounting Periods, Fiscal Year & Year-End Close



Figure 7: 12-Step Month-End Closing Procedure & Cryptographic SHA-256 Hash Chaining in LedgerCore Health — Flowing from Verified Reconciliations into a Tamper-Proof Permanent Ledger.

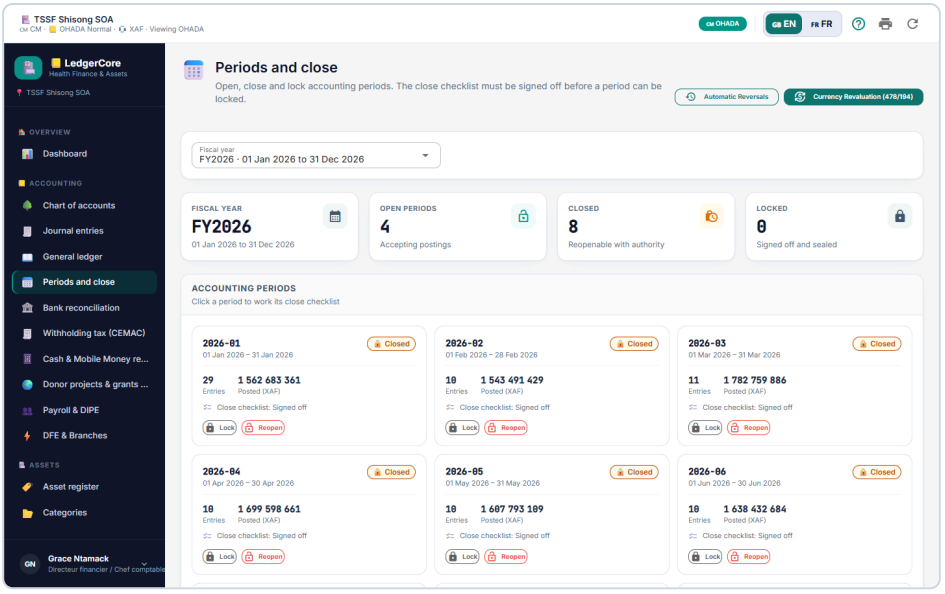


Figure 8: Accounting Period Management & 12-Step Close Interface — Showing Period Status (Open, Closed, Locked) and SHA-256 Hash Seal Integrity Verification.

7.1 Purpose & Statutory Cycle

Under **AUDCIF Article 13**, books must be closed periodically. LedgerCore implements a strict 3-tier lifecycle: - **Open:** Postings allowed by authorized clerks. - **Closed:** General postings blocked; adjustments require CFO override and audit logging. - **Locked:** Permanent immutable state sealed with a SHA-256 cryptographic hash.

7.2 Step-by-Step Procedure: Executing Month-End Closure

1. 🗓️ Navigate to **Accounting → Periods and close** (`/periods`).
2. 🔎 Locate the active period (e.g., `2026-09`) and click **Inspect Period**.
3. ✅ Complete the **12-Step Close Checklist**:
   - [x] Step 1: All draft entries verified and posted or deleted.
   - [x] Step 2: Bank and Mobile Money reconciliations completed with zero variance.
   - [x] Step 3: Cash desk registers audited and vault transfers verified.
   - [x] Step 4: Monthly asset depreciation run posted to Journal `AMO` .
   - [x] Step 5: Monthly payroll wage entry posted and reconciled against social declarations (DIPE).
   - [x] Step 6: Withholding taxes (WHT/TSR) reconciled and certificates generated.
   - [x] Step 7: Patient billing reconciled against pharmacy/ward consumption.
   - [x] Step 8: Foreign currency accounts revalued at month-end closing exchange rates.
   - [x] Step 9: Inter-branch reciprocal accounts (`181` / `185`) reconciled.
   - [x] Step 10: Physical inventory adjustments booked.
   - [x] Step 11: Departmental cost allocations reviewed.
   - [x] Step 12: Trial balance equilibrium confirmed.
4. 🔒 Click **Close Period**. Postings for the period are now locked to standard users.
5. 🔐 For year-end closure, click **Lock Period (Final Seal)** to generate the cryptographic hash chain.

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8. Module 6: Treasury, Bank Reconciliation & Cash Registers

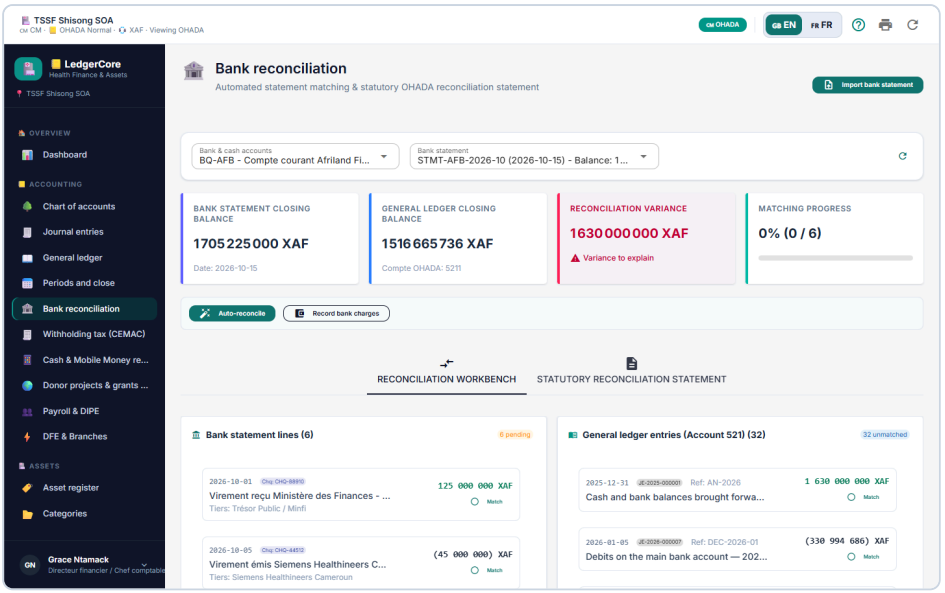


Figure 9: Treasury & Bank Reconciliation Workspace — Side-by-Side Electronic Statement Matching Against General Ledger Cash Books.

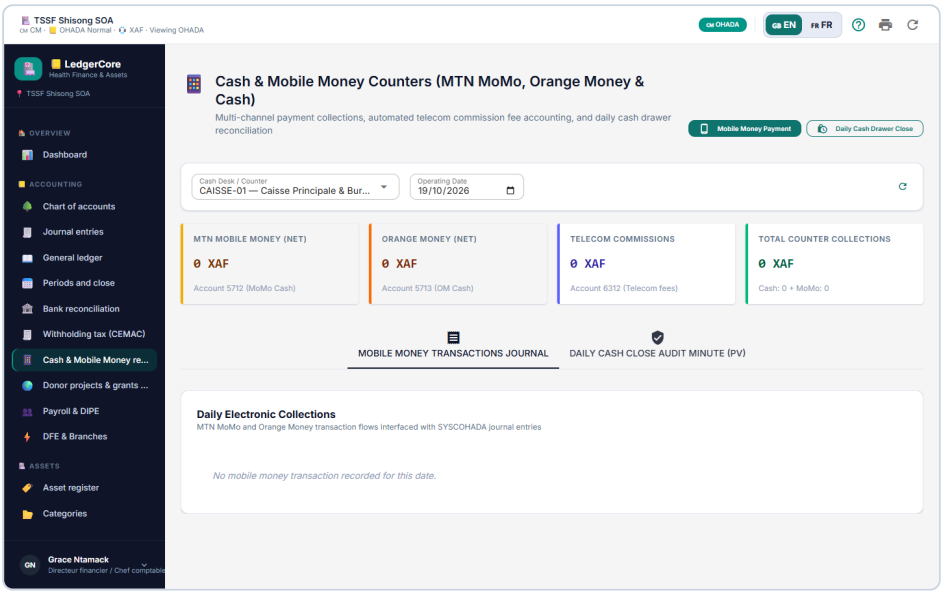


Figure 10: Cash Desk & Mobile Money Desk Register — Real-Time Denomination Balancing and Vault Transfer Reconciliation.

8.1 Step-by-Step Procedure: Reconciling a Bank Account

1. 🏦 Navigate to **Accounting → Bank reconciliation** (`/bank-reconciliation`).
2. 💳 In the **Bank Account** selector, choose the commercial bank account (e.g., `5211 — Commercial Bank Main Account`).
3. 📥 Click **Import Statement** and upload the electronic bank file (`MT940` , `OFX` , or standardized `CSV`).
4. 🔍 Review the side-by-side reconciliation table:
   - *Left column:* Bank statement lines.
   - *Right column:* General Ledger cash book lines.
5. ⚡ Click **Auto-Match**. Transactions with identical amounts, dates, and references are automatically reconciled.

6. ✍️ For unmatched bank charges (Account `631`) or interest fees, click **Post Adjustment** directly from the statement line to create the balancing voucher in Journal `BQ`.
7. 🎯 Once the **Unreconciled Variance** reaches `0.00 XAF`, click **Sign Off Reconciliation**.

8.2 Step-by-Step Procedure: Daily Cash Desk Balancing

1. 💵 Navigate to **Accounting → Cash & Mobile Money** (`/cash-registers`).
2. 🏧 Select the cash register (e.g., `Main Outpatient Cashier Desk`).
3. 🧮 Click **Count Cash / Denomination Sheet**.
4. 🔢 Enter physical bill and coin counts (10,000 XAF, 5,000 XAF, 2,000 XAF, etc.).
5. ⚖️ The system computes total counted cash against system recorded receipts.
6. 📝 If surplus or deficit exists, enter the explanation.
7. 🏦 Click **Confirm Count & Transfer to Vault** to generate the transfer voucher (Account `571` $\to$ `585`).

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9. Module 7: Fiscal Compliance & Withholding Tax (WHT / TSR)

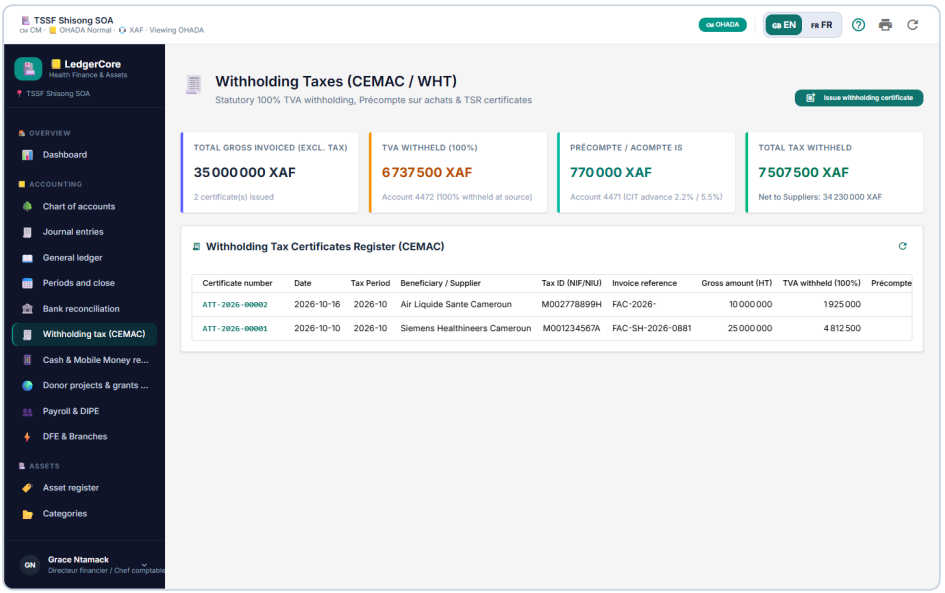


Figure 11: Withholding Tax (WHT / TSR) Register — Generating Statutory Certificates for Regional CEMAC Tax Filings.

9.1 Step-by-Step Procedure: Generating Withholding Tax Certificates

1. 📜 Navigate to **Accounting → Withholding tax (WHT)** (`/withholding-tax`).
2. 🗓️ Select the fiscal tax period (e.g., `September 2026`).
3. 📋 The table lists all supplier payments subject to statutory deductions:
   - **Special Income Tax on Services (TSR):** For international software licenses and foreign consulting.
   - **Advance Tax Withholding on Purchases:** For vendors lacking certified tax clearance certificates.
   - **VAT Withheld at Source:** Value-added tax deducted at disbursement.

4. ☑️ Check the box next to the supplier payment.
5. 🖨️ Click **Generate Statutory Certificate**.
6. 📥 Download and print the official signed certificate to accompany vendor payment and tax filing.

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10. Module 8: Donor-Funded Projects & Grant Accounting

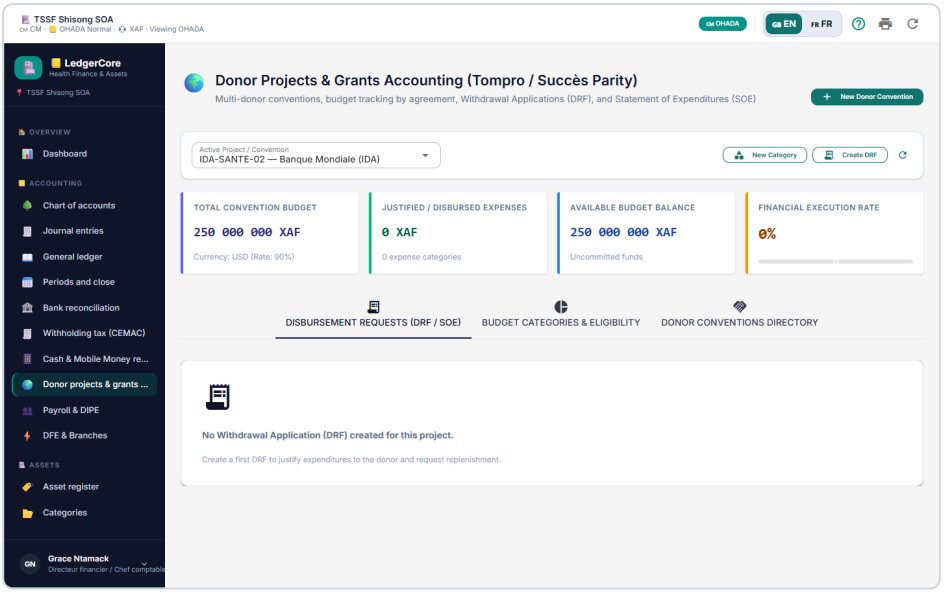


Figure 12: Donor Projects & Grants Management — Envelope Tracking, Real-Time Budget vs. Actuals, and Multi-Currency Burn Rates.

10.1 Step-by-Step Procedure: Creating and Tracking a Grant Budget

1. 🤝 Navigate to **Accounting → Donor Projects** (`/donor-projects`).
2. ➕ Click **+ New Project**.
3. 📝 Enter Project Code (e.g., `PRJ-GLOBALFUND-2026`), Title, Donor Agency, Grant Currency (USD/EUR), and Total Envelope.
4. 📊 Define approved line-item budget categories (Personnel, Medical Equipment, Consumables, Training).
5. 🏷️ **Tagging Expenses to Projects:** When capturing purchase vouchers in Journal `ACH` , enter the Project Code in the **Project** field on each relevant line.
6. 📈 **Monitoring Project Execution:** In the project overview, inspect the real-time **Budget vs. Actuals** comparison and burn rate percentage. Click **Export Donor Report** to generate the formatted financial summary.

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11. Module 9: Healthcare Payroll & Social Declarations (DIPE)

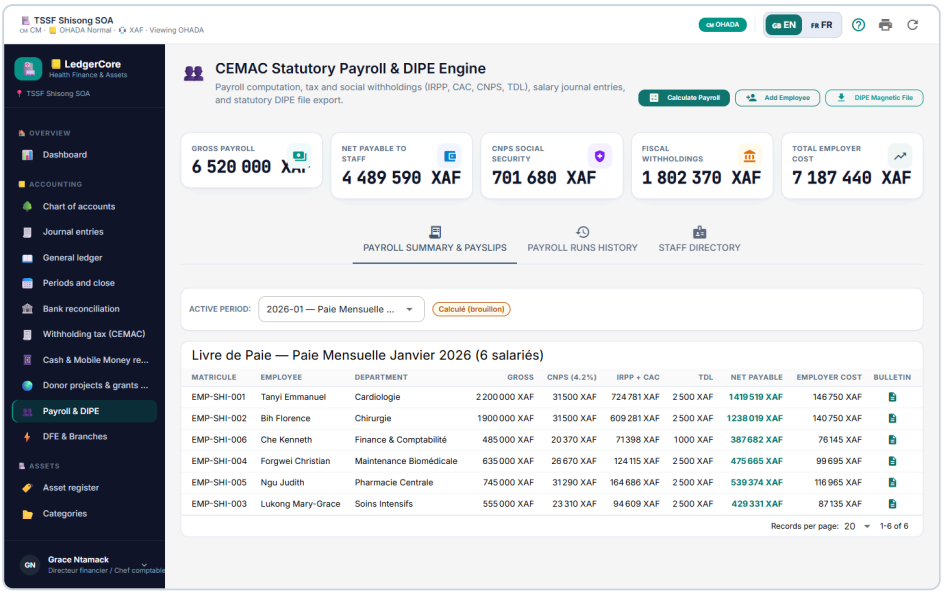


Figure 13: Healthcare Payroll Processing — Clinical Allowances, Statutory Social Security Deductions, Personal Income Tax, and DIPE Filing.

11.1 Step-by-Step Procedure: Monthly Payroll Processing & DIPE Generation

1. 👥 Navigate to **Accounting → Payroll & DIPE** (`/payroll`).
2. 🗓️ Select the payroll month (e.g., `September 2026`).
3. ⚙️ Click **Compute Payroll**. Review salary components:
   - Base Salaries
   - Technical Medical Allowances
   - On-Call & Night Duty Primes
   - Hazard Allowances
4. 🔍 Review statutory social and tax withholdings:
   - **Social Security (CNPS):** Employee share (4.2%) and employer share (family allowances, occupational risks, pension).
   - **Personal Income Tax (IRPP) & Surcharges (CAC):** Monthly statutory withholdings.
5. 📄 Click **Generate DIPE File**. The system compiles the statutory electronic format required for tax and social security electronic submission.
6. 🚀 Click **Post Wage Journal Entry**. LedgerCore automatically creates the balanced double-entry in Journal `PAI` (Debit Class 66 / Credit Class 42 & 43).

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12. Module 10: Patient Billing, Electronic Fee Schedules & Inter-Branch Transfers

12.1 Step-by-Step Procedure: Raising a Patient Billing Package

1. 🧾 Navigate to **Accounting → Patient Billing & Branches** (`/dfe-invoicing`).

2. ➕ Click **+ New Patient Invoice**.
3. 👤 Select Patient Name, Hospital File Number, and Ward.
4. 🩺 Add billed services from the standardized medical fee schedule (e.g., Surgery Package, Anesthesia, Hospitalization Bed Nights, Laboratory Tests).
5. 💳 **Configure Multi-Payer Split:**
   - In the **Payer Allocation** section:
     - Enter Patient Share (e.g., 20% copay).
     - Enter Insurer / Employer Mutual Fund Share (e.g., 80% coverage).
     - Select Third-Party Insurer account (e.g., `[CLI] Activa Health Insurance`).
6. 🖨️ Click **Issue Invoice**. The system prints the receipt and automatically creates the receivable entries in Journal `VTE`.

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13. Module 11: Enterprise Fixed Asset Register & Lifecycle Management

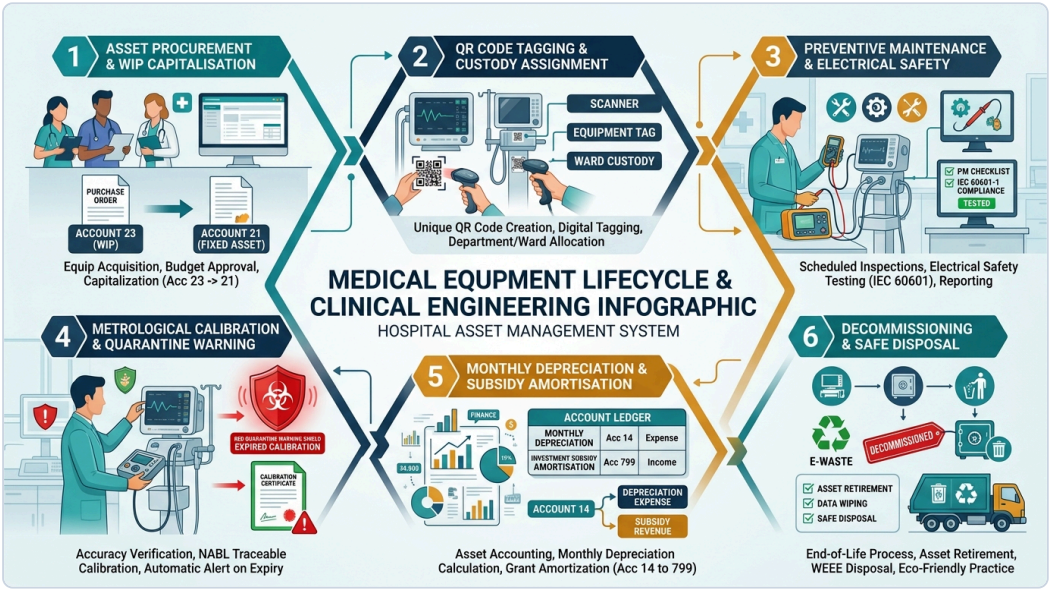


Figure 14: Medical Equipment Lifecycle & Clinical Engineering Infographic — 6 integrated stages: Procurement & WIP Capitalisation, QR Tagging, Maintenance & Electrical Safety, Calibration & Quarantine Warning, Depreciation & Subsidy Amortisation, and Decommissioning.

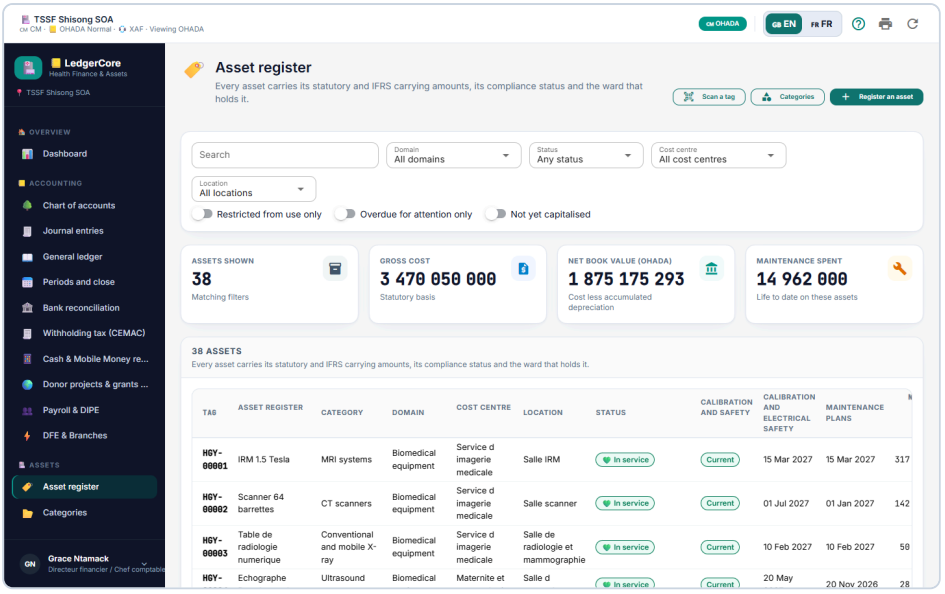


Figure 15: Enterprise Fixed Asset Register — Detailed Listing with Gross Book Value, Net Book Value (NBV), Ward Custodians, and Regulatory Calibration Indicators.

13.1 Step-by-Step Procedure: Registering New Medical Equipment

1. 🏷️ Navigate to **Assets → Asset register** (`/assets`).
2. ➕ Click **+ Register Asset**.
3. 📝 **Fill General Attributes:**
   - **Asset Name:** Official equipment description (e.g., `Hamilton-C3 Intensive Care Ventilator`).
   - **Domain:** Select `Biomedical` .
   - **Category:** Choose category (e.g., `Medical & Surgical Equipment`).
   - **Serial Number & Model:** Manufacturer serial number and model designation.
4. 💰 **Valuation & Accounting Parameters:**
   - **Acquisition Cost:** Invoice gross cost in XAF.
   - **Commissioning Date:** Date equipment entered clinical service.
   - **Depreciation Method:** Straight-line or Degressive (Declining Balance).
   - **Useful Lifetime:** Set operational life (e.g., 5 years / 60 months).
5. 📍 **Location & Custody Assignment:**
   - **Department / Ward:** Assign to clinical unit (e.g., `Intensive Care Unit`).
   - **Room:** Room/bed location (e.g., `ICU Bay 2`).
   - **Custodian:** Assign supervising ward head.
6. 🖨️ **Save & Print QR Tag:**
   - Click **Save Asset**.
   - Click **Print QR Tag** to generate the waterproof barcode sticker to affix to the machine chassis.

13.2 Step-by-Step Procedure: Capitalizing Work-in-Progress (WIP Transfer 23 -> 21)

1. 🏗️ In the Asset Register, click the **Capitalize WIP** button.
2. 📋 Select the accumulated WIP cost lines sitting in Account `231` (Work-in-Progress Capital Assets).

3. 🎯 Specify the final destination asset account (e.g., `2151 — Complex Technical Installations`).
4. 🚀 Click **Execute Capitalization**. The system registers the completed physical asset and posts the balancing journal entry in Journal `IMM` (`Debit 2151 / Credit 231`).

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14. Module 12: Asset Depreciation & Leases (IFRS 16)

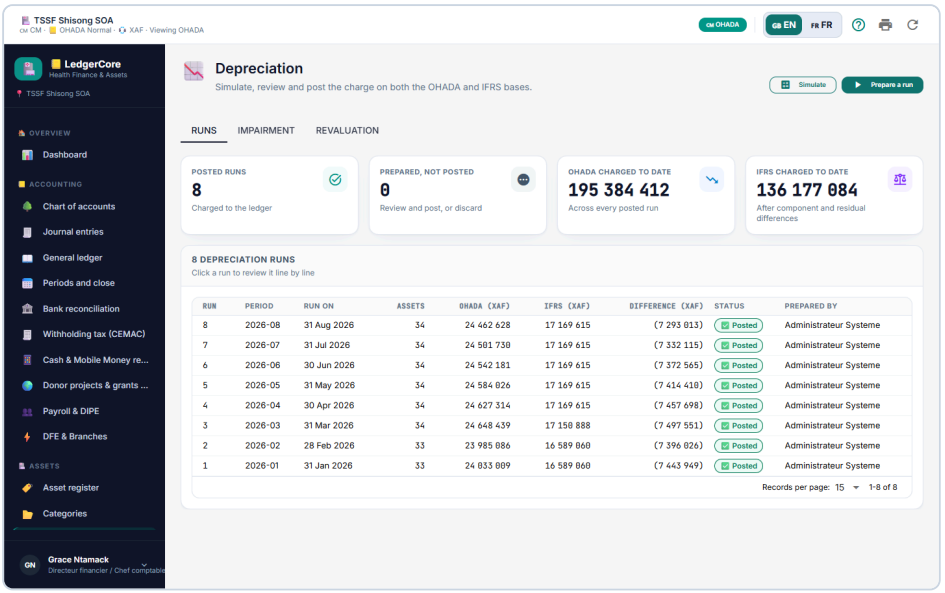


Figure 16: Asset Depreciation Workspace — Dual-Method Simulation (Straight-Line & Degressive) with Journal AMO Automated Depreciation Charges.

14.1 Step-by-Step Procedure: Executing a Monthly Depreciation Run

1. 📉 Navigate to **Assets → Depreciation** (`/depreciation`).
2. ⚙️ Click **Run Depreciation**.
3. 🗓️ Select the period (e.g., `September 2026`).
4. 🔍 Click **Simulate**. The engine computes depreciation charges line-by-line across all active assets under both OHADA and IFRS rules.
5. 📊 Review the simulation table:
   - *Statutory OHADA Depreciation Charge:* Debits Account `681` / Credits Account `28x`.
   - *IFRS Depreciation:* Evaluated on economic useful life.
   - *IFRS Delta:* Variance between statutory and international depreciation.
   - *Subsidy Amortization:* Write-back of donor subsidies (Debit Account `14` / Credit Account `799`).
6. 🚀 Click **Post Run**. LedgerCore generates and commits the automated entry in Journal `AMO`.

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15. Module 13: Physical Asset Inventory & Barcode Verification

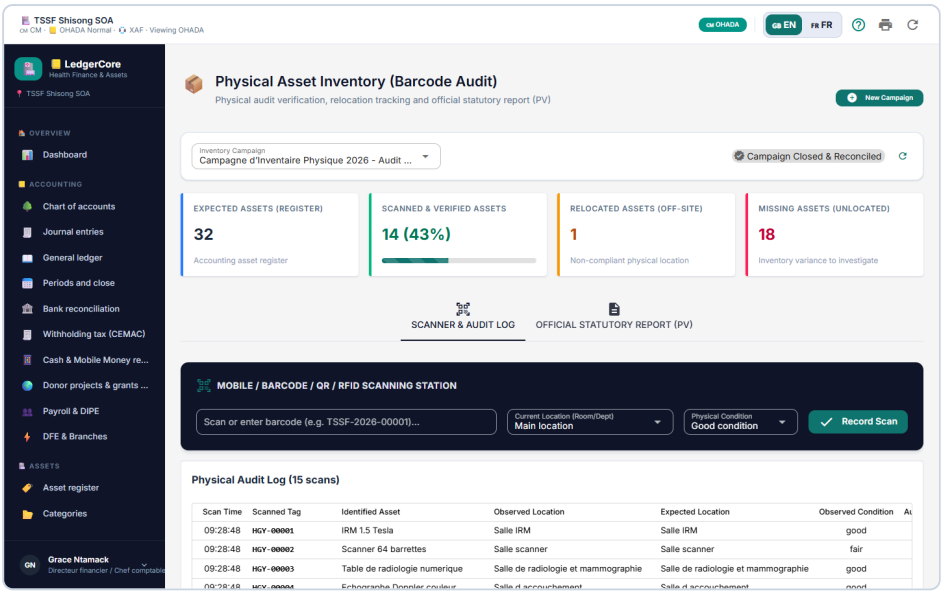


Figure 17: Physical Inventory & Barcode Sweep — Auditing Assets via Mobile Barcode Scanners with Variance Reconciliation.

15.1 Step-by-Step Procedure: Conducting a Physical Asset Audit

1. 🧮 Navigate to **Assets → Physical inventory** (`/asset-inventory`).
2. ➕ Click **+ New Campaign** (e.g., `Annual Inventory 2026 - Surgery & ICU`).
3. 📍 Set the target building and asset domain.
4. 📷 **Conduct Field Auditing:** Open the web application on a mobile tablet; scan the QR code on each physical machine. The system checks off the asset as **Verified** and records timestamp and coordinates.
5. 📊 **Review Variance Analysis:**
   - **Verified:** Machine present in its assigned location.
   - **Displaced:** Machine found in a different ward than registered. Click **Reassign Location** to update custody.
   - **Missing:** Equipment not scanned during the sweep. Flag for clinical investigation or write-off.
6. 🏁 Click **Finalize Campaign** to publish the audit report.

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16. Module 14: Hospital CMMS, Work Orders & Calibration (IEC 60601)

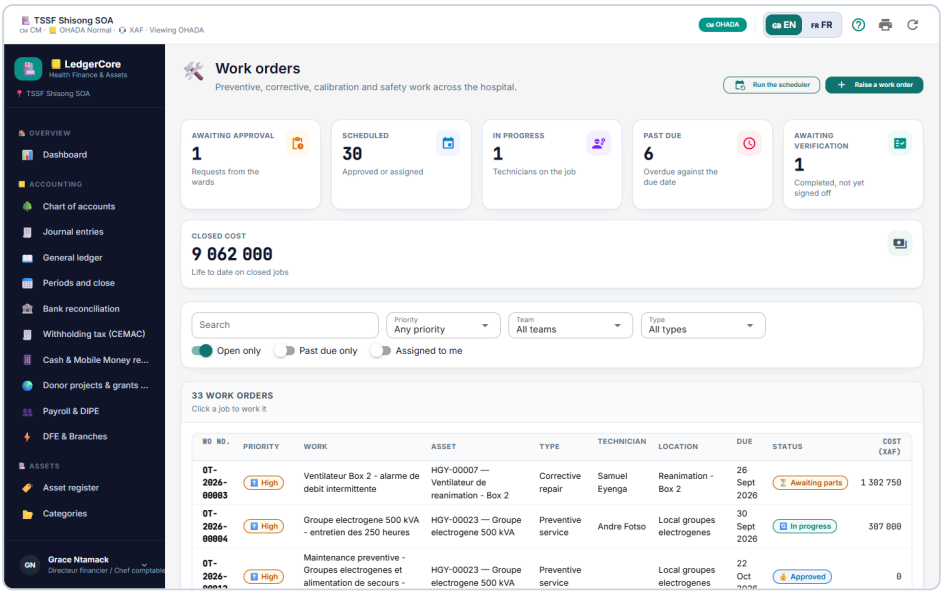


Figure 18: Clinical CMMS Work Orders Pipeline — Managing Corrective Repairs, Preventive Cycles, and Technician Field Assignments.

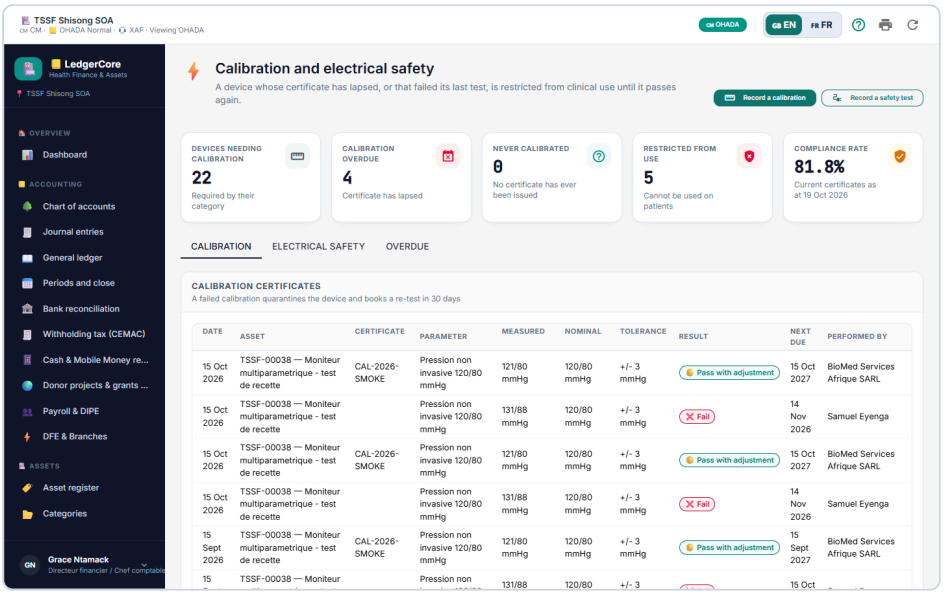


Figure 19: Biomedical Calibration & Electrical Safety Dashboard — Monitoring Certificate Validity, Test Results, and Patient Safety Quarantine Enforcement.

16.1 Step-by-Step Procedure: Raising, Executing & Closing a Work Order

1. 🚨 **Raising a Breakdown Request (Ward Staff):**
   - Click **Maintenance → Work orders** (`/maintenance`). Click **+ Raise Work Order**.
   - Select the affected Asset Tag (e.g., `AST-BIO-2026-0042`).
   - Describe the clinical fault (e.g., `Infusion pump alarming error E04 - occlusion sensor failure`).
   - Set urgency: *Critical*, *High*, *Medium*, *Low*. Click **Submit Request**.

2. 👨‍⚕️ **Triaging & Assignment (Biomedical Manager):**
   - In the work order queue, open the ticket under `Awaiting Approval` .
   - Assign to a Biomedical Technician (e.g., `samuel.eyenga@hgy.cm`). Click **Approve & Dispatch**.

3. 🛠️ **Execution & Parts Consumption (Technician):**
   - Technician opens ticket on mobile device and clicks **Start Job**.
   - Under **Parts Used**, click **+ Add Part**. Select parts consumed from warehouse stock (e.g., `Pressure Sensor Membrane`).
   - Enter technician diagnostic notes.
4. 🩺 **Clinical Verification & Closeout:** Ward nurse or doctor tests functionality and signs on screen. Biomedical manager verifies electrical safety and clicks **Verify & Close**.

16.2 Step-by-Step Procedure: Logging IEC 60601 Safety & Calibration Tests

1. 🔬 In the asset card, select the **Compliance & Safety** tab.
2. ⚡ Under **IEC 60601-1 Electrical Safety Test**, enter measured readings:
   - Protective Earth Resistance ($\le 0.2\ \Omega$)
   - Chassis Leakage Current ($\le 500\ \mu\text{A}$)
   - Patient Lead Leakage ($\le 10\ \mu\text{A}$)
3. 📜 Under **Calibration**, enter the test date, issuing laboratory, certificate number, and expiry date. Attach the PDF certificate.
4. 🚨 **Automatic Quarantine Behavior:** If an electrical safety reading fails, or if the calibration expiry date passes, LedgerCore instantly sets the asset's status to **Quarantined**. A prominent red warning appears on the clinical dashboard prohibiting the machine from patient use until re-tested and passed.

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17. Module 15: General Services, Errands & Procurement Logistics

17.1 Step-by-Step Procedure: Executing Errand Runs & Shopping Lists

1. 🛒 Navigate to **General Services → Shopping list** (`/shopping`) or **Errands** (`/errands`).
2. 📋 General Services supervisor creates an errand run (e.g., `Procurement of Diesel for Standby Generator`).
3. 📱 Driver / Logistics officer opens the task on their mobile phone.
4. 💵 At the supply depot, check off each acquired item and enter the unit purchase price.
5. 📷 Tap **Attach Receipt** to snap a photo of the cash receipt or delivery note using the device camera.
6. 🏁 Tap **Complete Errand**. The system routes the expense voucher to accounting for petty cash reimbursement.

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18. Module 16: Mobile Technician Worklist

18.1 Step-by-Step Procedure: Using the Touch-Friendly Mobile Interface

1. 📱 Log into LedgerCore on a mobile device as a technician (e.g., `samuel.eyenga@hgy.cm`).
2. 📋 Navigate to **Maintenance → My worklist** (`/worklist`).
3. 🎯 The screen displays prioritized tasks scheduled for today.
4. 📷 Tap **Scan Asset** to activate the camera; scan the equipment QR tag to immediately view service manual PDFs and historical repair logs.
5. ⏱️ Tap **Start Timer** to track active wrench time.
6. ✍️ When work is completed, tap **Sign & Complete** to capture digital clinical sign-off.

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19. Module 17: Financial Statements, Annual Reporting & Tax Packet

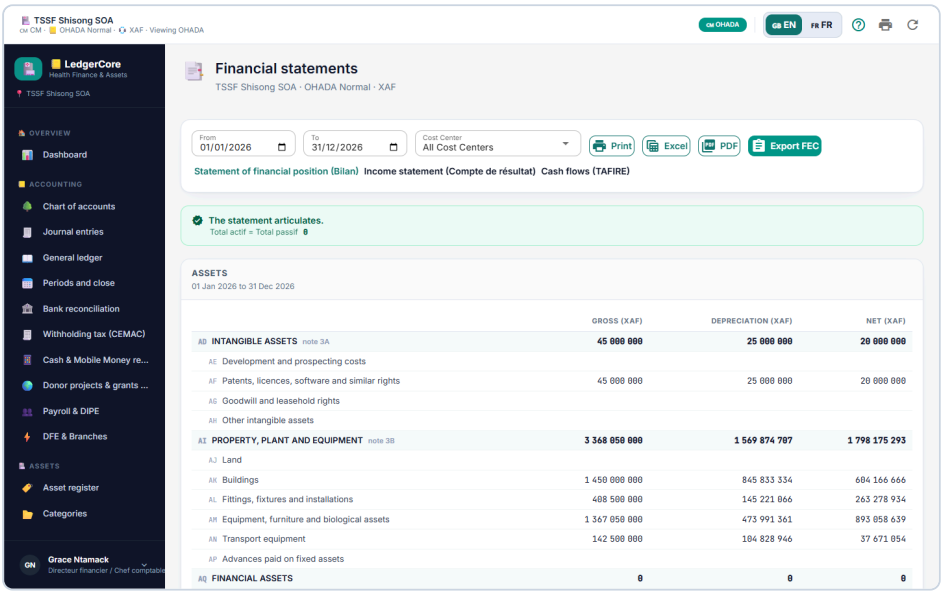


Figure 20: Financial Statements & Statutory Reports — SYSCOHADA Balance Sheet, Income Statement, Statement of Cash Flows, and IFRS Dual-Standard Bridge.

19.1 Step-by-Step Procedure: Generating Statutory Financial Statements

1. 📊 Navigate to **Reporting → Financial statements** (`/reports`).
2. 🗓️ Select the target fiscal period and standard (**OHADA** or **IFRS**).
3. 📒 **OHADA Presentation:**
   - **Balance Sheet:** Fixed Assets, Current Assets, Cash & Cash Equivalents, Equity & Net Capital Funds, Long-Term Liabilities & Debt, Short-Term Bank Overdrafts.
   - **Income Statement:** Hospital Production, Operating Turnover, Value Added, Gross Operating Surplus (EBITDA), Operating Income, Financial Profit/Loss, Extraordinary Result, Net Surplus/Deficit.
   - **Statement of Cash Flows (TAFIRE):** Cash flow from operating, investing, and financing activities.

4. 🌐 **IFRS Presentation:** Statement of Financial Position, Statement of Profit or Loss, Statement of Cash Flows, Statement of Changes in Equity (SOCE).
5. 🌉 **Inspecting the Dual-Standard Bridge:** Click the **Dual-Standard Bridge** tab. The table displays: $\text{OHADA Net Result} + \sum \text{IFRS Adjustments} = \text{IFRS Net Profit}$.
6. 📄 **Exporting the Accounting Audit File (FEC):** Click **Export FEC**. The system downloads the 18-column dematerialized journal file compliant with CEMAC tax inspection standards.

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20. Module 18: Security, Governance & Administration

20.1 Step-by-Step Procedure: Provisioning Users and Inspecting Audit Logs

1. ⚙️ Navigate to **Administration → Admin console** (`/admin`).
2. 👤 **User Creation:** Click **+ Add User**. Enter user details, email, and assign role permissions (`admin` , `cfo` , `accountant` , `asset_manager` , `biomed_tech` , `services_tech` , `ward_head`).
3. 🕵️ **Audit Trail Inspection:** Click **Reporting → Action history** (`/audit`). Every transaction creation, posting, modification, deletion, and reversal is logged with timestamp, user ID, client IP address, and before/after JSON payloads.
4. 🔐 **Verifying Cryptographic Seal:** Under **Audit Seal Verification**, run the automated scanner. The system re-computes SHA-256 hashes across all journal entries to confirm zero manual database tampering.

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21. Standard Operating Procedures (SOPs) & Practical Workflows

SOP 1: Supplier Invoice Reception, 3-Way Match & Voucher Entry

1. 📦 Procurement receives supplier delivery note and vendor invoice.
2. 🔍 Verify delivery note against physical warehouse goods received note.
3. 💻 Accounting clerk opens LedgerCore: **Accounting → Journal Entries → + New Entry**.
4. 🛍️ Select Journal `ACH` . Enter supplier invoice number in **Reference ***.
5. ✍️ Impute expense account (Class 6), input cost centre, and credit auxiliary supplier subledger (Account `401100` , partner selected).
6. 💾 Save as Draft for Chief Accountant verification.

SOP 2: Clinical Equipment Breakdown to Return-to-Service Protocol

1. 🚨 Nurse notices ventilator failure; navigates to **Maintenance → Work orders → + Raise Work Order**.
2. 🏷️ Selects equipment tag and marks priority *Critical*.
3. 👨‍⚕️ Biomedical manager receives instant dashboard notification; assigns ticket to technician.
4. 🛠️ Technician arrives in ward, scans asset QR tag, replaces sensor, and executes IEC 60601 electrical leakage test.

5. ⚡ Technician logs safety test results: status updates from *Quarantined* to *Operational*.
6. ✍️ Clinical head inspects machine, signs on mobile screen, and technician marks ticket *Completed*.

LedgerCore Health — Engineered for Statutory Excellence, Clinical Safety, and Operational Integrity.